Lattice Boltzmann Theory

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1 Introduction

The Lattice Boltzmann Method is a numerical technique for the simulation of fluid flows that originates from the kinetic theory of gases rather than from a direct discretization of the Navier–Stokes equations (NSE). The Boltzmann equation, which describes the evolution of a single-particle distribution function on a *mesoscopic* scale.

It can be shown that, under suitable assumptions, a solution of the Boltzmann equation recovers a solution of the Navier–Stokes equations for the same physical problem and its numerical scheme turns out to be remarkably simple, local, and well suited to parallel computing. This is the main reason why the LBM has become a popular alternative to traditional Computational Fluid Dynamics (CFD) solvers.

1.1 Physical and Mathematical Model

The LBM is derived from the continuous Boltzmann kinetic equation through a sequence of discretization steps in velocity space, physical space, and time. The macroscopic fluid behavior emerges from the dynamics thanks to a Chapman–Enskog multiscale expansion, which recovers the incompressible Navier–Stokes equations in the low-Mach-number limit.
The collision process is modeled using two opterators: the Bhatnagar–Gross–Krook (BGK) operator and two-relaxation-time (TRT) operator.

The BGK is a single-relaxation-time approximation that drives the distribution function towards a local equilibrium. The relaxation time τ controls the kinematic viscosity of the simulated fluid through the relation

$$\nu = c_s^2 \left(\tau - \frac{1}{2} \right), \tag{1}$$

where c_s^2 is the lattice speed of sound. For the D2Q9 lattice, $c_s^2 = 1/3$, so the inverse relaxation time (the collision frequency) is computed as

$$\frac{1}{\tau} = \frac{2}{6\nu + 1}. \tag{2}$$

The TRT operator decomposes every pair in symmetric and anti-symmetric parts, the first relaxation controls the physical viscosity,

$$\tau_+ = \frac{1}{2} + 3\nu, \tag{3}$$

while the antisymmetric relaxation time is fixed

$$\tau_- = \frac{1}{2} + \frac{1/4}{\tau_+ - 1/2}, \tag{4}$$

The main advantage of TRT over BGK is that the physical viscosity, governed by τ_+, is decoupled from the relaxation of the non-hydrodynamic component. This extra degree of freedom is particularly useful in the presence of bounce-back walls, where it can reduce the sensitivity of the effective wall location and improve numerical stability, especially at low viscosity.

BGK and TRT share the same equilibrium distribution, streaming procedure, macroscopic reconstruction and boundary-condition code.

1.2 Stability Considerations

Numerical stability of the BGK scheme requires $\tau > \frac{1}{2}$.
As the Reynolds number Re increases, the kinematic viscosity ν decreases and τ approaches the stability limit. As a consequence:

- high-Reynolds-number simulations require a sufficiently fine grid resolution;
- the lid velocity must remain small in lattice units, so as to keep the simulation in the low-Mach-number, weakly compressible regime;
- density fluctuations must remain small.

Stability is also tightly linked to the local density field. In the routine that computes the macroscopic moments (`compute_rho_u`), the density and momentum are accumulated as

$$\rho \mathrel{+}= f_i, \qquad u_x \mathrel{+}= c_{ix}\, f_i, \qquad u_y \mathrel{+}= c_{iy}\, f_i, \tag{5}$$

and the macroscopic velocity is finally recovered by dividing by the density,

$$u = \frac{u_x}{\rho}, \qquad v = \frac{u_y}{\rho}. \tag{6}$$

If ρ becomes too small at any lattice node, the computed velocity grows, triggering a numerical instability.

2 Model Description

The core quantity of the LBM is the *discrete-velocity distribution function* $f_i(\mathbf{x}, t)$, often referred to as the particle *population* associated with the i-th discrete velocity direction. This function describes the state of the fluid by tracking how particles collide with one another and stream between neighboring lattice cells.

The discretized Lattice Boltzmann equation in velocity and time reads

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t,\ t + \Delta t) = f_i(\mathbf{x}, t) + \Omega_i(\mathbf{x}, t), \tag{7}$$

where $\mathbf{c}_i$ is the discrete microscopic velocity associated with direction i, and Ω_i is the collision operator. Eq. (7) naturally splits the algorithm into two conceptually distinct sub-steps: *streaming* (the left-hand side, propagation of populations to neighboring nodes) and *collision* (the right-hand side, local relaxation towards equilibrium).

As mentioned above, the collision operator adopted in this work is the BGK operator,

$$\Omega_i(f) = -\frac{f_i - f_i^{\mathrm{eq}}}{\tau}\, \Delta t, \tag{8}$$

with τ the relaxation time and Δt the simulation time step.

The equilibrium distribution f_i^{eq}, computed explicitly in the routines `init_equilibrium` and `collide`, is obtained from a second-order expansion of the Maxwell–Boltzmann distribution in the macroscopic velocity:

$$f_i^{\mathrm{eq}}(\mathbf{x}, t) = w_i\, \rho \left[1 + \frac{\mathbf{u} \cdot \mathbf{c}_i}{c_s^2} + \frac{(\mathbf{u} \cdot \mathbf{c}_i)^2}{2c_s^4} - \frac{\mathbf{u} \cdot \mathbf{u}}{2c_s^2} \right], \tag{9}$$

where:

- w_i is the lattice weight associated with direction i, and ρ is the local fluid density;
- $\mathbf{c}_i$ is the set of discrete particle velocities of the chosen lattice;
- $\mathbf{u}$ is the macroscopic fluid velocity;
- $c_s^2 = \frac{1}{3}\,\Delta x^2/\Delta t^2$ is the lattice speed of sound, which sets the relation between pressure and density in the (weakly compressible, isothermal) LBM.

Note: the value of c_s^2 does not need to be computed explicitly at run time, since in lattice units ($\Delta x = \Delta t = 1$) it reduces to the constant 1/3.

The D2Q9 Velocity Set
The project employs the standard two-dimensional, nine-velocity (D2Q9) lattice, consisting of: 1 rest population ($\mathbf{c}_0 = \mathbf{0}$), 4 axis-aligned populations; and 4 diagonal populations.
In lattice units ($\Delta x = \Delta t = 1$), the discrete velocities $\mathbf{c}_i = (c_{ix}, c_{iy})$ and the associated lattice weights w_i.

For the D2Q9 lattice, the lattice speed of sound satisfies $c_s^2 = 1/3$ (in lattice units), a value that guarantees second-order isotropy of the lattice and the correct correspondence between the LBM and the incompressible Navier–Stokes equations at the macroscopic level.

3 Boundary Conditions

Boundary nodes are tagged in the code through a boundary mask, with each node assigned one of the following types: rigid-wall bounce-back, moving-wall bounce-back, periodic, pressure-periodic inlet, and pressure-periodic outlet.

3.1 Rigid-Wall Bounce-Back

A no-slip stationary wall is modeled with bounce-back: a population that would stream in from outside the domain is replaced by the population traveling in the opposite direction,

$$f_i \leftarrow f_{\bar{i}}. \tag{10}$$

This reverses the microscopic momentum normal and tangential to the wall and enforces a no-slip condition at the macroscopic level, to the accuracy of the bounce-back formulation. Rigid-wall bounce-back is used at the lower wall of the Couette problem, the upper and lower walls of the Poiseuille channel, and the left, right, and bottom walls of the lid-driven cavity.

3.2 Moving-Wall Bounce-Back

For a wall moving with velocity $\mathbf{u}_w$, the reflected population requires a momentum correction,

$$f_i = f_{\bar{i}} - 2w_{\bar{i}}\rho\,\frac{\mathbf{c}_{\bar{i}} \cdot \mathbf{u}_w}{c_s^2}. \tag{11}$$

Since $c_s^{-2} = 3$ for D2Q9, this correction injects the required wall momentum directly into the fluid. This condition is used at the upper plate of the Couette problem and at the moving lid of the cavity, with the reference wall velocity supplied through `init_vel` (typically $\mathbf{u}_w = (0.1, 0)$).

3.3 Periodic Boundaries

For a standard periodic boundary, the source location is wrapped to the opposite side of the domain,

$$\mathbf{x}_w = (\mathbf{x} - \mathbf{c}_i) \bmod \mathbf{N}, \qquad f_i(\mathbf{x}) = f_i(\mathbf{x}_w), \tag{12}$$

with $\mathbf{N}$ the lattice dimensions. This condition is applied in the streamwise direction of the Couette-flow simulation, where the flow is statistically homogeneous along x.

3.4 Pressure-Periodic Boundaries

Pressure-driven (Poiseuille) flow requires a periodic boundary with an imposed pressure-level difference between the two ends of the domain. The incoming population is reconstructed as

$$f_i(\mathbf{x}) = f_i^{\mathrm{eq}}(\rho_b, \mathbf{u}_w) + f_i(\mathbf{x}_w) - f_i^{\mathrm{eq}}(\rho_w, \mathbf{u}_w), \tag{13}$$

where $\mathbf{x}_w$ is the wrapped node on the opposite side of the domain and ρ_b is the prescribed inlet or outlet equilibrium density level. In the code these levels are named `pin` and `pout`; since they enter Eq. (13) in the position occupied by density, they should be interpreted as density-like pressure parameters, related to actual pressure through the equation of state, up to the constant factor c_s^2.

4 Flow Configurations

Three flow configurations are used to exercise, in increasing order of complexity, the collision operators and the boundary-condition set described above.

Plane Couette Flow

Plane Couette flow consists of a viscous fluid confined between two parallel plates a distance H apart: the lower plate is stationary, the upper plate moves at constant velocity U_w. For a steady, fully developed flow with no streamwise pressure gradient, the x-momentum equation reduces to $\nu\, d^2u_x/dy^2 = 0$, which, with $u_x(0) = 0$ and $u_x(H) = U_w$, gives the linear profile

$$u_x(y) = U_w\, \frac{y}{H}. \tag{14}$$

In the implementation, this problem combines a rigid bounce-back lower wall, a moving bounce-back upper wall, and periodic left/right boundaries, and is therefore a direct test of the moving-wall correction (Eq. (11)) together with periodic streaming.

Plane Poiseuille Flow

Plane Poiseuille flow describes pressure-driven flow between two stationary parallel plates. For a steady, fully developed flow, the streamwise momentum equation reduces to

$$0 = -\frac{1}{\rho}\frac{dp}{dx} + \nu\frac{d^2u_x}{dy^2}, \tag{15}$$

which, with no-slip at $y = 0, H$, yields the parabolic profile

$$\frac{u_x}{U_{\max}} = 4\,\frac{y}{H}\left(1 - \frac{y}{H}\right), \qquad U_{\max} = \frac{-1}{8\rho\nu}\frac{dp}{dx}H^2. \tag{16}$$

For a channel of length L, the corresponding pressure drop is $\Delta p = 8\rho\nu U_{\max}L/H^2$. In the implementation this case uses rigid bounce-back at the upper and lower walls together with the pressure-periodic condition of Eq. (13) at the inlet and outlet, and is therefore the natural test of the pressure-periodic boundary treatment against a known analytical shape.

Lid-Driven Cavity

The lid-driven cavity encloses the fluid in a square domain whose bottom and side walls are stationary while the top wall translates horizontally at constant velocity U_{lid},

$$\mathbf{u}(x, 0) = \mathbf{u}(0, y) = \mathbf{u}(L, y) = \mathbf{0}, \qquad \mathbf{u}(x, L) = (U_{\mathrm{lid}}, 0), \tag{17}$$

with $Re = U_{\mathrm{lid}}L/\nu$. Unlike Couette and Poiseuille flow, the cavity problem is genuinely two-dimensional and admits no simple closed-form solution: the moving lid drives a dominant re-circulating vortex whose shape, and whose secondary corner vortices, evolve with the Reynolds number. It combines rigid bounce-back on three sides with the moving-wall condition on the lid, and thus exercises the full boundary-condition set together with the collision operator in a genuinely two-dimensional setting; quantitative assessment of this case against reference data is left to the benchmarking stage.

5 The Time-Stepping Algorithm

Each simulation time step consists of a fixed cyclic sequence of operations:

1. perform the streaming step;
2. apply the boundary conditions;
3. compute the macroscopic moments $\rho(\mathbf{x}, t)$ and $\mathbf{u}(\mathbf{x}, t)$ from $f_i(\mathbf{x}, t)$;
4. compute the equilibrium distribution $f_i^{\mathrm{eq}}(\mathbf{x}, t)$;
5. apply the collision operator;
6. write the macroscopic fields to output;
7. advance the time step and check for convergence.

5.1 Key Parameters

The simulation is governed by the following physical and numerical parameters:

- Reynolds number: Re;
- Kinematic viscosity: $\nu = \dfrac{u_{\mathrm{lid}}\, L_x}{Re}$;
- Relaxation time: $\tau = 3\nu + \frac{1}{2}$.

References

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[2] U. Ghia, K. N. Ghia, C. T. Shin, "High-Re solutions for incompressible flow using the Navier-Stokes equations and a multigrid method," *Journal of Computational Physics*, vol. 48, no. 3, pp. 387–411, 1982.